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System and method of performing a read operation

Abstract

A memory device includes a memory array having memory cells, a page buffer coupled to the memory array through bit lines and including a latch, and a control logic coupled to the page buffer. The control logic is configured to perform a first read operation on the memory cells, and perform a second read operation on the memory cells by dividing the memory cells into a plurality of groups of memory cells, and performing a second read operation on the plurality of groups of memory cells based on a first set of develop times. Each of the first set of develop times is different. The control logic is also configured to perform a third read operation on the memory cells by performing a third read operation on the plurality of groups of memory cells based on a second set of develop times. Each of the second set of develop times is different. The control logic is further configured to determine a read develop time based on the third read operation.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10090046	12/2017	Park et al.	N/A	N/A
2017/0133087	12/2016	Park	N/A	N/A
2019/0096479	12/2018	Yu	N/A	N/A
2020/0051645	12/2019	Choo	N/A	G11C 16/28
2020/0286545	12/2019	Yoon	N/A	N/A
2021/0005265	12/2020	Lee	N/A	N/A
2021/0005268	12/2020	Kim	N/A	G11C 11/5671
2022/0057968	12/2021	Kim	N/A	N/A
2022/0093184	12/2021	Kim et al.	N/A	N/A
2023/0251781	12/2022	Son	N/A	N/A
2024/0028218	12/2023	Jeong	N/A	N/A
2024/0078038	12/2023	Li	N/A	G06F 3/0679

OTHER PUBLICATIONS

International Search Report directed to International Patent Application No. PCT/CN2023/075962, mailed Jul. 18, 2023, 3 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/184,312, filed on Mar. 15, 2023, which is a continuation of International Application No. PCT/CN2023/075962, filed on Feb. 14, 2023, both of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(1) This description generally relates to the field of semiconductor technology, and more particularly, to a system and method for performing a read operation in a NAND memory.

BACKGROUND

(2) As memory devices are shrinking to smaller die size to reduce manufacturing cost and increase storage density, scaling of planar memory cells faces challenges due to process technology limitations and reliability issues. For example, a three-dimensional (3D) memory architecture can address the density and performance limitation in planar memory cells.

(3) In a NAND flash memory, many layers of memory cells can be stacked vertically such that storage density per unit area can be greatly increased. The vertically stacked memory cells can form memory strings, where the channels of the memory cells are connected in each memory string. Each memory cell can be addressed through a word line and a bit line. Data (i.e., logic states) of the memory cells in an entire memory page sharing the same word line can be read or programmed simultaneously. However, due to aggressive scaling, reliability can be a concern for a NAND flash memory.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated herein and form part of the specification, illustrate the present disclosure and, together with the description, further serve to explain the principles of the disclosure and enable a person of skill in the relevant art(s) to make and use the disclosure.

(2) FIGS. 1 and 2A-2B illustrate a storage system with one or more memory chips, according to some embodiments.

(3) FIG. 3 illustrates a schematic diagram of a memory device, according to some embodiments.

(4) FIG. 4 illustrates a schematic of a memory device, according to some embodiments.

(5) FIG. 5 illustrates a schematic diagram of an exemplary threshold voltage distribution of a memory device, according to some embodiments.

(6) FIG. 6A illustrates an example schematic circuit diagram of a memory device, according to some embodiments.

(7) FIG. 6B illustrates an example schematic diagram of a page buffer of a memory device, according to some embodiments.

(8) FIG. 7 illustrates an exemplary signal waveform illustrating a voltage level of a sensing node of FIG. 6B, according to some embodiments.

(9) FIG. 8A illustrates an example method **800** of performing a read operation on a memory device, according to some embodiments.

(10) FIG. 8B illustrates an example threshold voltage distribution and data structures during the process flow of FIG. 8A, according to some embodiments.

(11) The features and advantages of the present disclosure will become more apparent from the detailed description set forth below when taken in conjunction with the drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

(12) Aspects of the present disclosure will be described with reference to the accompanying drawings.

DETAILED DESCRIPTION

(13) Although specific configurations and arrangements are discussed, it should be understood that this is done for illustrative purposes only. A person skilled in the pertinent art will recognize that other configurations and arrangements can be used without departing from the spirit and scope of

the present disclosure. It will be apparent to a person skilled in the pertinent art that the present disclosure can also be employed in a variety of other applications.

(14) It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “some embodiments,” etc., indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of a person skilled in the pertinent art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

(15) In general, terminology can be understood at least in part from usage in context. For example, the term “one or more” as used herein, depending at least in part upon context, can be used to describe any feature, structure, or characteristic in a singular sense or can be used to describe combinations of features, structures, or characteristics in a plural sense. Similarly, terms, such as “a,” “an,” or “the,” again, can be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. In addition, the term “based on” can be understood as not necessarily intended to convey an exclusive set of factors and can, instead, allow for existence of additional factors not necessarily expressly described, again, depending at least in part on context.

(16) As used herein, the term “substrate” refers to a material onto which subsequent material layers are added. The substrate includes a “top” surface and a “bottom” surface. The top surface of the substrate is typically where a semiconductor device is formed, and therefore the semiconductor device is formed at a top side of the substrate unless stated otherwise. The bottom surface is opposite to the top surface and therefore a bottom side of the substrate is opposite to the top side of the substrate. The substrate itself can be patterned. Materials added on top of the substrate can be patterned or can remain unpatterned. Furthermore, the substrate can include a wide array of semiconductor materials, such as silicon, germanium, gallium arsenide, indium phosphide, etc. Alternatively, the substrate can be made from an electrically non-conductive material, such as a glass, a plastic, or a sapphire wafer.

(17) As used herein, the term “layer” refers to a material portion including a region with a thickness. A layer has a top side and a bottom side where the bottom side of the layer is relatively close to the substrate and the top side is relatively away from the substrate. A layer can extend over the entirety of an underlying or overlying structure, or can have an extent less than the extent of an underlying or overlying structure. Further, a layer can be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer can be located between any set of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer can extend horizontally, vertically, and/or along a tapered surface. A substrate can be a layer, can include one or more layers therein, and/or can have one or more layer thereupon, there above, and/or there below. A layer can include multiple layers. For example, an interconnect layer can include one or more conductive and contact layers (in which contacts, interconnect lines, and/or vertical interconnect accesses (VIAs) are formed) and one or more dielectric layers.

(18) As used herein, the term “nominal/nominally” refers to a desired, or target, value of a characteristic or parameter for a component or a process step, set during the design phase of a product or a process, together with a range of values above and/or below the desired value. The range of values can be due to slight variations in manufacturing processes or tolerances. As used herein, the terms “about” or “approximately” indicate the value of a given quantity that can vary based on a particular technology node associated with the subject semiconductor device. Based on the particular technology node, the terms “about” or “approximately” can indicate a value of a given quantity that varies within, for example, 10-30% of the value (e.g., +10%, +20%, or +30% of

the value).

(19) Currently, in memory devices, especially in high density memory devices, threshold voltage (V_t) distribution shift can be impacted by many factors, such as programmed cells charge loss with over time, noises, long NAND's service life, etc., thus is a common and critical problem. After V_t distribution shift, the pre-defined read level cannot track V_t distribution, thereby causing read fails. A system and method for performing a read operation in a NAND memory, to track V_t distribution is needed.

(20) FIG. 1 illustrates a block diagram of an electronic device **S1** having a storage system **10**, according to some embodiments. In some embodiments, the electronic device **S1** can be a mobile phone, a desktop computer, a laptop computer, a tablet, a vehicle computer, a gaming console, a printer, a positioning device, a wearable electronic device, a smart sensor, a virtual reality (VR) device, an argument reality (AR) device, or any other suitable electronic devices having storage therein. Storage system **10** (e.g., a NAND storage system) can include a memory controller **20** and one or more semiconductor memory devices **25-1**, **25-2**, **25-3**, . . . , **25-n**. Each semiconductor memory device **25** (hereafter just "memory device") can be a NAND device (e.g., "flash," "NAND flash" or "NAND"). Storage system **10** can communicate with a host **15** through memory controller **20**, where memory controller **20** can be connected to one or more memory chips **25-1**, **25-2**, **25-3**, . . . , **25-n**, via one or more memory channels **30-1**, **30-2**, **30-3**, . . . , **30-n**. In some embodiments, each memory device **25** can be managed by memory controller **20** via one or more memory channels **30-1**, **30-2**, **30-3**, . . . , **30-n**.

(21) In some embodiments, host **15** can include a processor of an electronic device, such as a central processing unit (CPU), or a system-on-chip (SoC), such as an application processor (AP). Host **15** can send data to be stored at storage system **10** and/or can retrieve data from stored in storage system **10**.

(22) In some embodiments, memory controller **20** can handle I/O requests received from host **15**, ensure data integrity and efficient storage, and manage memory device **25**. To perform these tasks, memory controller **20** can run firmware **21**, which can be executed by one or more processors **22** (e.g., micro-controller units, CPU) of memory controller **20**. For example, memory controller **20** can run firmware **21** to map logical addresses (e.g., address utilized by the host associated with host data) to physical addresses in memory device **25** (e.g., actual locations where the data is stored). Memory controller **20** also runs firmware **21** to manage defective memory blocks in the memory device **25**, where the firmware **21** can remap the logical address to a different physical address, i.e., move the data to a different physical address. Memory controller **20** can also include one or more memories **23** (e.g., DRAM, SRAM, EPROM, etc.), which can be used to store various metadata used by the firmware **21**. In some embodiments, the memory controller **20** can also perform error recovery through an error correction code (ECC) engine **29**. ECC is used to detect and correct the raw bit errors that occur within each memory device **25**.

(23) In some embodiments, the memory channels **30** can provide data and control communication between the memory controller **20** and each memory device **25** via a data bus. The memory controller **20** can select one of the memory device **25** according to a chip enable signal.

(24) In some embodiments, each memory device **25** in FIG. 1 can include one or more memory devices **100**, where each memory device can be a NAND memory.

(25) In some embodiments, memory controller **20** and one or more memory device **25** can be integrated into various types of storage devices, for example, be included in the same package, such as a universal Flash storage (UFS) package or an cMMC package. That is, storage system **10** can be implemented and packaged into different types of end electronic products. In one example as shown in FIG. 2A, memory controller **20** and a single memory device **25** can be integrated into a memory card **26**. Memory card **26** can include a PC card (PCMCIA, personal computer memory card international association), a CF card, a smart media (SM) card, a memory stick, a multimedia card (MMC, RS-MMC, MMCmicro), an SD card (SD, miniSD, microSD, SDHC), a UFS, etc.

Memory card **26** can further include a memory card connector **24** coupling memory card **26** with a host (e.g., the host **15** in FIG. **1**). In another example as shown in FIG. **2B**, memory controller **20** and multiple memory devices **25** can be integrated into a solid state drive (SSD) **27**. SSD **27** can further include a SSD connector **28** coupling SSD **27** with a host (e.g., the host **15** in FIG. **1**).

(26) FIG. **3** illustrates a top-down view of a memory device **100**, according to some embodiments. The example configuration shown in FIG. **3** is given as a non-limiting example and it is to be appreciated that memory is scalable. In some embodiments, memory device **100** can include one or more memory planes **101**, each of which can include a plurality of memory blocks **103**. Identical and concurrent operations can take place at each memory plane **101**. Memory block **103**, which can be megabytes (MB) in size, is the smallest size to carry out erase operations. Memory device **100** can include, for example, four memory planes **101**. Each memory plane **101** can include, for example, six memory blocks **103**. Each memory block **103** can include a plurality of memory cells, where each memory cell can be addressed through interconnections such as bit lines and word lines. The bit lines and word lines can be laid out perpendicularly (e.g., in rows and columns, respectively), forming an array of metal lines. The direction of bit lines and word lines are labeled as “BL” and “WL” in FIG. **3**.

(27) In some embodiments, memory device **100** can also include a periphery region **105**, an area surrounding memory planes **101**. The periphery region **105** can include many digital, analog, and/or mixed-signal circuits to support functions of the memory device, for example, page buffers, row and column decoders and sense amplifiers. Peripheral circuits use active and/or passive semiconductor devices, such as transistors, diodes, capacitors, resistors, etc., as would be apparent to a person of ordinary skill in the art.

(28) In some embodiments, the arrangement of the memory planes **101** in the memory device **100** and the arrangement of the memory blocks **103** in each memory plane **101** illustrated in FIG. **3** are only used as an example, which does not limit the scope of the present disclosure.

(29) FIG. **4** illustrates a schematic diagram of the memory device **100**, according to some embodiments. In some embodiments, memory device **100** can include one or more memory blocks **103** (e.g., **103-1**, **103-2**, **103-3**). Each memory block **103** can include a plurality of memory strings **212**. Each memory string **212** includes a plurality of memory cells **340**. Memory cells **340** sharing the same word line forms a memory page **432**. Memory string **212** can also include at least one field effect transistor (e.g., MOSFET) at each end, which is controlled by a lower select gate (“LSG”) **332** and a top select gate (“TSG”) **334**, respectively. Lower select gates (“LSGs”) can also be referred to as bottom select gates (“BSGs”). The drain terminal of the top select transistor **334-T** can be connected to a bit line **341**, and the source terminal of the lower select transistor **332-T** can be connected to an array common source (“ACS”) **430**. ACS **430** can be shared by the memory strings **212** in an entire memory block, and is also referred to as the common source line.

(30) In some embodiments, memory device **100** can also include a periphery circuit that can include many digital, analog, and/or mixed-signal circuits to support functions of the memory block **103**, for example, a page buffer **52**, a row decoder/word line driver **40**, a column decoder/bit line driver **50**, a controller **70**, a voltage generator **65** and an input/output buffer **55**. Controller **70** can include one or more control circuits. In some aspects, controller **70** can include one or more registers, buffers, and/or memories to store one or more trim settings as described in the present disclosure. These circuits can include active and/or passive semiconductor devices, such as transistors, diodes, capacitors, resistors, etc., as would be apparent to a person of ordinary skill in the art.

(31) In some embodiments, memory blocks **103** can be coupled with the row decoder/word line driver **40** via word lines (“WLs”) **333**, lower select gates (“LSGs”) **332** and top select gates (“TSGs”) **334**. Memory blocks **103** can be coupled with page buffer **52** via bit lines (“BLs”) **341**. Row decoder/word line driver **40** can select one of the memory blocks **103** on the memory device **100** in response to an X-path control signal provided by the controller **70**. Row decoder/word line

driver **40** can transfer voltages provided from the voltage generator **65** to the word lines according to the X-path control signal. During the read and program operation, the row decoder/word line driver **40** can transfer a read voltage $V_{sub.read}$ and a program voltage $V_{sub.pgm}$ to a selected word line and a pass voltage $V_{sub.pass}$ to an unselected word line according to the X-path control signal received from the controller **70**.

(32) In some embodiments, column decoder/bit line driver **50** can transfer an inhibit voltage $V_{sub.inhibit}$ to an unselected bit line and connect a selected bit line to ground according to a Y-path control signal received from controller **70**. In the other words, column decoder/bit line driver **50** can be configured to select or unselect one or more memory strings **212** according to the Y-path control signal from controller **70**. The page buffer **52** can be configured to read and program (write) data from and to the memory block **103** according to the control signal Y-path control from the controller **70**. For example, the page buffer **52** can store one page of data to be programmed into one memory page **432**. In another example, page buffer **52** can perform verify operations to ensure that the data has been properly programmed into each memory cell **340**. In yet another example, during a read operation, page buffer **52** can sense current flowing through the bit line **341** that reflects the logic state (i.e., data) of the memory cell **340** and amplify small signal to a measurable magnification.

(33) In some embodiments, in order to increase the efficiency of a write operation, column decoder/bit line driver **50** can transfer a bias voltage $V_{sub.bias}$ to a selected bit line according to a Y-path control signal from controller **70** and the data to be programmed from page buffer **52**.

(34) In some embodiments, input/output buffer **55** can transfer the I/O data from/to the page buffer **52** as well as addresses ADDR or commands CMD to the controller **70**. In some embodiments, input/output buffer **55** can function as an interface between memory controller **20** (in FIG. 1) and memory device **100** on memory device **25**.

(35) In some embodiments, controller **70** can control page buffer **52** and row decoder/word line driver **40** in response to the commands CMD transferred by the input/output buffer **55**. During the program operation, controller **70** can control row decoder/word line driver **40** and page buffer **52** to program a selected memory cell. During the read operation, controller **70** can control row decoder/word line driver **40** and the page buffer **52** to read a selected memory cell. The X-path control signal and the Y-path control signal include a row address X-ADDR and a column address Y-ADDR that can be used to locate the selected memory cell in the memory block **103**. The row address X-ADDR can include a page index PD, a block index BD and a plane index PL to identify memory page **432**, memory block **103**, and memory plane **101** (in FIG. 3), respectively. The column address Y-ADDR can identify a byte or a word in the data of the memory page **432**.

(36) In some embodiments, voltage generator **65** can generate voltages to be supplied to word lines and bit lines under the control of controller **70**. The voltages generated by voltage generator **65** include the read voltage $V_{sub.read}$, the program voltage $V_{sub.pgm}$, the pass voltage $V_{sub.pass}$, the inhibit voltage $V_{sub.inhibit}$, the bit line bias voltage $V_{sub.bias}$, etc.

(37) It is noted that the arrangement of the electronic components in the storage system **10** and the memory device **100** in FIGS. 1, 2A-2B, and 3-4 are shown as non-limiting examples. In some embodiments, storage system **10** and memory device **100** can have other layout and can include additional components. Components (e.g., controller **70**, I/O buffer **55**) on memory device **100** shown in FIG. 4 can also be moved off memory device **100**, as a stand-alone electric component in the storage system **10**. Components (e.g., controller **70**, I/O buffer **55**) on memory device **100** shown in FIG. 4 can also be moved to other components in storage system **10**, for example, a portion of controller **70** can be combined with memory controller **20** and vice versa.

(38) FIG. 5 illustrates a schematic diagram of an exemplary threshold voltage distribution of a memory device, according to some embodiments.

(39) Referring back to FIG. 4, in some embodiments, memory block **103** can be formed based on floating gate technology. In some embodiments, the memory block **103** can be formed based on

charge trapping technology. The NAND flash memory based on charge trapping can provide high storage density and high intrinsic reliability. Storage data in the form of logic states (“states,” e.g., threshold voltages V_{th} of the memory cell **340**) depends on the number of charge carriers trapped in the memory film of the memory cell **340**.

(40) In some embodiments, in a NAND flash memory, a read operation and a write operation (also referred to as program operation) can be performed for the memory page **432**, and an erase operation can be performed for the memory block **103**.

(41) In some embodiments, in a NAND memory, the memory cell **340** can be in an erased state ER or a programmed state P1. Initially, memory cell **340** in memory block **103** can be reset to the erased state ER as logic “1” by implementing a negative voltage difference between control gates **333** and channel such that trapped charge carriers in the memory film of memory cells **340** can be removed. For example, the negative voltage difference can be induced by setting control gates **333** of memory cells **340** to ground, and applying a high positive voltage (an erase voltage V_{erase}) to ACS **430**. At the erased state ER (“state ER”), the threshold voltage V_{th} of memory cells **340** can be reset to the lowest value.

(42) In some embodiments, during programming (i.e., writing), a positive voltage difference between control gates **333** and channel can be established by, for example, applying a program voltage V_{pgm} (e.g., a positive voltage pulse between 10 V and 20 V) on control gate **333**, and grounding the corresponding bit line **341**. As a result, charge carriers (e.g., electrons) can be injected into the memory film of memory cell **340**, thereby increasing the threshold voltage V_{th} of memory cell **340**. Accordingly, memory cell **340** can be programmed to the programmed state P1 (“state P1” or logic “0”).

(43) In some embodiments, the state of the memory cell (e.g., state ER or state P1) can be determined by measuring or sensing the threshold voltage V_{th} of the memory cell. During a read operation, a read voltage V_{read} can be applied on control gate **333** of the memory cell and current flowing through the memory cell can be measured at bit line **341**. A pass voltage V_{pass} can be applied on unselected word lines to switch on unselected memory cells.

(44) In some embodiments, a NAND flash memory can be configured to operate in a single-level cell (SLC) mode. To increase storage capacity, a NAND flash memory can also be configured to operate in a multi-level cell (MLC) mode, a triple-level cell (TLC) mode, a quad-level cell (QLC) mode, or a combination of any of these modes. In the SLC mode, a memory cell stores 1 bit and has two logic states, logic {1 and 0}, i.e., states ER and S1. In the MLC mode, a memory cell stores 2 bits, and has four logic states, logic {11, 10, 01, and 00}, i.e., states ER, M1, M2, and M3. In the TLC mode, a memory cell stores 3 bits, and has eight logic states, logic {111, 110, 101, 100, 011, 010, 001, 000}, i.e., states ER, and states T1-T7. In the QLC mode, a memory cell stores 4 bits and has 16 logic states, logic {1111, 1110, 1101, 1100, 1011, 1010, 1001, 1000, 0111, 0110, 0101, 0100, 0011, 0010, 0001, 0000}, i.e., states ER, and states Q1-Q15. Memory controller **20** of storage system **10** (see FIG. 1) can convert data received from host **15** into corresponding logic states of the memory cells on memory devices **100** and vice versa.

(45) In some aspects of the QLC mode, states P1-P15 corresponds to states Q1-Q15. In some aspects, each state of the memory cells can correspond to a specific range of threshold voltage $V_{sub.th}$, where the threshold voltage $V_{sub.th}$ distribution of each state can be represented by a probability density. In some aspects, the states other than the erased state ER can be programmed by using an incremental step pulse programming (ISPP) scheme where the programming voltage $V_{sub.pgm}$ can be incrementally increased by adding a step pulse $V_{sub.step}$. For example, the QLC states can be programmed from state ER with a lower threshold voltage to state Q15 with a highest threshold voltage.

(46) In some aspects, after programming, states P1-P15 can be verified by using one or more pre-defined read reference voltages, during a verification process. By applying one or more of the pre-defined read reference voltages to the control gate of a target memory cell, the range of the memory

cell's threshold voltage $V_{sub.th}$ can be determined.

(47) For example, to verify if a memory cell is at state ER, the read reference voltage $V_{sub.R1}$ can be used. If the target memory cell is at state ER, the threshold voltage $V_{sub.th}$ of the target memory cell is lower than the read reference voltage $V_{sub.R1}$. The target memory cell can be switched on and form a conductive path in the channel. If the target memory cell is at any one of the states P1-P15, the threshold voltage $V_{sub.th}$ of the target memory cell is higher than the pre-defined read reference voltage $V_{sub.R1}$. The target memory cell is thereby switched off. By measuring or sensing the current through the target memory cell at the corresponding bit line, via the page buffer 52, the threshold voltage $V_{sub.th}$ or the state of the target memory cell can be verified.

(48) In some aspects, as shown in diagram 510, to determine the states ER and P1-P15 for the QLC mode, the pre-defined read reference voltages, for example, including, $V_{sub.R1}$, $V_{sub.R13}$, $V_{sub.R14}$ and $V_{sub.R15}$, can be used. For example, in the QLC mode, the threshold voltage of state ER is below $V_{sub.R1}$, and the threshold voltage of state P15 is above $V_{sub.R15}$, where the threshold voltages of state P14 is between $V_{sub.R14}$ and $V_{sub.R15}$ and the threshold voltages of state P13 is between $V_{sub.R13}$ and $V_{sub.R14}$.

(49) In some aspects, as shown in diagram 520, the threshold voltage $V_{sub.th}$ distribution of each state of P1-P15 may shift over time. As shown in diagram 520, the change of a threshold voltage distribution of memory cells may be differently illustrated according to a programmed state. For example, in the case of a lower program state P1, P2 and P3 a distribution may tend to shift to the slight right side. Moreover, in the case of upper program states P12, P13, P14 and P15, the distribution may tend to spread in the slight left side. By applying one or more of the pre-defined read reference voltages, including for example, $V_{sub.R1}$, $V_{sub.R13}$, $V_{sub.R14}$ and $V_{sub.R15}$, to the control gate of the target memory cell, the range of the memory cell's threshold voltage V_{th} may not be determined appropriately. In some aspects, a valley may include an intersection between a first threshold voltage distribution of a first state and a second threshold voltage distribution of a second state. The second state may include an adjacent or a neighboring state of the first state. For example, valley 530 may include an intersection between a threshold voltage distribution of a program state P14 and a threshold voltage distribution of a program state P15.

(50) In memory devices, especially in high density memory devices, threshold voltage (V_t) distribution shift can be impacted by many factors, such as programmed cells charge loss with over time, noises, long NAND's service life, etc., thus is a common and critical problem. After V_t distribution shift, the pre-defined read level cannot track V_t distribution, thereby causing read fails.

(51) FIG. 6A illustrates an example schematic circuit diagram 600 of a memory device, according to aspects of the present disclosure. The example schematic circuit diagram 600 includes a memory block 604 and a peripheral circuit 602. In some aspects, the example schematic circuit diagram 600 includes a plurality of memory strings 610, 620 and 630, with each of memory strings 610, 620 and 630 having a plurality of memory cells 608. The memory string 610 also includes at least one field effect transistor (e.g., MOSFET) at each end, which is controlled by a lower select gate (LSG) 612 and a top select gate (TSG) 614, respectively. The memory cell 608 can be controlled by a control gate, where the control gate can be connected to a word line 640 of the example schematic circuit diagram 600. The drain terminal of the TSG 614 can be connected to the bit line 618, and the source terminal of the LSG 612 can be connected to an array common source (ACS) 616. The ACS 616 can be shared by the memory strings 610 in an entire memory block, and is also referred to as the common source line. In some examples, memory string 620 can be associated with bit line 628 and memory string 630 can be associated with bit line 638.

(52) In some aspects, the example schematic circuit diagram 600 can be formed based on the floating gate technology. In some aspects, the example schematic circuit diagram 600 can be formed based on charge trapping technology. The NAND flash memory based on charge trapping can provide high storage density and high intrinsic reliability. Storage data or logic states (e.g.,

threshold voltage $V_{sub.th}$ of the memory cell **608**) depends on the amount of charge trapped in a storage layer. In some aspects, the memory block **604** can be a three-dimensional (3D) memory device, and the example schematic circuit diagram **600** can be a 3D memory array, where the memory cells **608** can be vertically stacked on top of each other.

(53) In a NAND memory, the memory cell **608** can be in an erase state ER or a programmed state P1. Initially, all memory cells **608** in the example schematic circuit diagram **600** can be reset to the erase state ER as logic “1” by implementing a negative voltage difference between control gates and source terminals of the memory cells (e.g., the array common source **616**) such that all the trapped electronic charges in the storage layer of the memory cells **608** can be removed. For example, the negative voltage difference can be induced by setting the control gates of the memory cells **608** to ground, and applying a high positive voltage to the array common source **616**. At the erase state ER (“state ER”), the threshold voltage $V_{sub.th}$ of the memory cells **608** can be reset to the lowest value, and can be measured or sensed at the bit line **618**.

(54) During programming (i.e., writing), a programming voltage $V_{sub.pgm}$ (e.g., a positive voltage pulse between 10 V and 20 V) can be applied on the control gate such that electronic charges (e.g., electrons) can be injected into the storage layer of the memory cell **608**, and thereby increase the threshold voltage $V_{sub.th}$ of the memory cell **608**. Thus, the memory cell **608** is programmed to the state P1. In some examples, memory cell **608** may be programmed to different states, such as P2-P15 in QLC mode.

(55) FIG. **6B** illustrates an example schematic diagram of a page buffer of a memory device, according to some embodiments. FIG. **6B** can be described with regard to page buffer **52** of FIG. **4** and elements of FIGS. **1-6A**. Not all of the depicted components may be used, however, and one or more implementations may include additional components not shown in the figure. Variations in the arrangement and types of the components may be made without departing from the spirit or scope of the claims as set forth herein. Additional, different or fewer components may be provided.

(56) As illustrated in FIG. **6B**, page buffer **52** includes a first page buffer structure **658**, a second page buffer structure **668**, a third page buffer structure **678**, and a latch **680**. Latch **680** is connected with first page buffer structure **658**, second page buffer structure **668**, and third page buffer structure **678** to perform a read operation of a memory device (e.g., memory device **100**). In some examples, first page buffer structure **658** may be connected with memory string **610** through bit line **618** of FIG. **6A**. In some examples, second page buffer structure **668** may be connected with memory string **620** through bit line **628** of FIG. **6A**. In some examples, third page buffer structure **678** may be connected with memory string **630** through bit line **638** of FIG. **6A**.

(57) As shown in FIG. **6B**, first page buffer structure **658** includes a sensing node (SO) **650**, a pre-charge path **652**, a SO discharge path **654** and a sense latch **656**. Second page buffer structure **668** includes a sensing node (SO) **660**, a pre-charge path **662**, a SO discharge path **664** and a sense latch **666**. Third page buffer structure **678** includes a sensing node (SO) **670**, a pre-charge path **672**, a SO discharge path **674** and a sense latch **676**.

(58) As described with reference to FIG. **4**, during a read operation, the row decoder/word line driver **40** can transfer a read voltage V_{read} to a selected word line to an unselected word line according to the X-path control signal received from the controller **70**. The page buffer **52** can be configured to read data from and to the memory block **103** according to the control signal Y-path control from the controller **70**. During the read operation, page buffer **52** can sense current flowing through the bit line **341** that reflects the logic state (i.e., data) of the memory cell **340** and amplify small signal to a measurable magnification.

(59) During a read operation, first page buffer structure **658** may precharge bit line **618** through pre-charge path **652** by a control logic (e.g., control circuit **70**) and may sense at SO **650** whether a selected memory cell is turned on or off. Second page buffer structure **668** and third page buffer structure **678** may respectively precharge bit lines connected thereto based on the same procedure as the precharging operation of the above-described first page buffer structure **658**.

(60) During a read operation, bit line **618** and SO **650** may be precharged to a pre-determined level during a pre-charge period. A current can be generated (e.g., in a channel) so as to flow into first page buffer structure **658** through the bit line **618**. During a develop period with a develop time, when the selected memory cell is in a first state (e.g., on cell), a charge charged at SO **650** may be discharged to the array common source (ACS) **616** through bit line **618** and the channel of a cell string through SO discharge path **654**. In this case, because the current flowing to first page buffer structure **658** is relatively great, the speed of a voltage drop of SO **650** may be relatively fast. On the other hand, during the develop period with the develop time, when the selected memory cell is in a second state (e.g., off cell), it may be difficult for a charge charged at SO **650** to be discharged to ACS **616** through bit line **618**. Accordingly, because the current flowing to first page buffer structure **658** is relatively small, the speed of a voltage drop of SO **650** may be relatively slow.

(61) During a latch period, a state of SO **650** associated with the selected memory cell may be latched to sense latch **656** as a first state of the selected memory cell with a logical value of 1. Alternatively, a state of SO **650** associated with the selected memory cell may be latched to sense latch **656** as a second state of the selected memory cell with a logical value of 0. In some examples, a threshold voltage of the selected memory cell in the first state is larger than a read voltage in the read operation and a threshold voltage of the selected memory cell in the second state is smaller than a read voltage in the read operation. In some examples, a memory cell is in the first state when the memory cell is turned on when a read voltage level is applied, and a memory cell is in the second state when the memory cell is turned off when a read voltage level is applied.

(62) In some examples, second page buffer structure **668** and third page buffer structure **678** may respectively discharge bit lines connected thereto and sense a state of sensing node as a first state or a second state of the selected memory cell based on the same procedure as the operations of the above-described first page buffer structure **658**.

(63) After the latch period, latch **680** may store the data associated with each of sense latch **656**, sense latch **666**, and sense latch **676**. Latch **680** may perform one or more modification on the stored data associated with each of sense latch **656**, sense latch **666**, and sense latch **676**. In some examples, latch **680** may modify the logical value of the data associated with each of sense latch **656**, sense latch **666**, and sense latch **676**. For example, latch **680** may modify the logical value of 1 to the logical value of 0. Alternatively, latch **680** may modify the logical value of 0 to the logical value of 1.

(64) In some examples, during a read operation, a develop time during the develop period may be different for each of first page buffer structure **658**, second page buffer structure **668** and third page buffer structure **678**. The sensing node, pre-charge path, SO discharge path and sense latch associated with each of first page buffer structure **658**, second page buffer structure **668** and third page buffer structure **678** can be controlled separately in different configurations.

(65) FIG. 7 illustrates an exemplary signal waveform illustrating a voltage level of a sensing node of FIG. 6B, according to some embodiments. FIG. 7 can be described with regard to sensing node **650**, **660** or **670** of FIG. 6 and elements of FIGS. 1-6B. Not all of the depicted components may be used, however, and one or more implementations may include additional components not shown in the figure. Variations in the arrangement and types of the components may be made without departing from the spirit or scope of the claims as set forth herein. Additional, different or fewer components may be provided.

(66) Referring to FIG. 7, the level change of the sensing node SO (e.g., sensing node **650**, **660** or **670**) according to a threshold voltage level of a memory cell and a latch result according to a develop period may be briefly illustrated. A time period **710** from time T0 to time T1 may be referred to as the precharge period as discussed with reference to FIG. 6B, a time period **720** from time T1 to time T2 may be referred to as a develop period as discussed with reference to FIG. 6B. During the develop period, the change of a voltage level of the sensing node SO about a strong on cell may correspond to a curve C1. The strong on cell or a strong off cell may not be significantly

affected by the slight change of the develop time. The voltage change of the sensing node SO for sensing memory cells may correspond to curve C2, and a threshold voltage of each of the memory cells may be placed near a read voltage. The curve C2 illustrates a develop slope of a memory cell having a threshold voltage slightly lower than the read voltage. A time period 730 between time T2 and T3 may be referred to as the sense period as discussed with reference to FIG. 6B.

(67) In some examples, in the case of a memory cell having a threshold voltage that is relatively higher than a read voltage, the level change of the sensing node SO may be relatively small. In the case of a memory cell having a threshold voltage that is relatively lower than a read voltage, the level change of the sensing node SO may be relatively great. In some aspects, memory cells with threshold voltages distributed around a valley between a first threshold voltage distribution of a first state and a second threshold voltage distribution of a second state (e.g., valley 530) may be memory cells placed at a boundary between an on cell and an off cell. Accordingly, a distinction between an on cell and an off cell about the memory cells may be changed according to the develop time. That is, even though the develop time slightly decreases, each of the memory cells having threshold voltages distributed around a valley may be determined as an “off cell”. On the other hand, even though the develop time slightly increases, each of the memory cells having threshold voltages distributed around a valley may be determined as an “on cell”. That is, in memory cells having a threshold voltage level similar to a read voltage level to be provided to a word line, a sensing operation may be performed the same as a sensing operation using a read voltage increased, by reducing a develop time. On the other hand, in memory cells having a threshold voltage level similar to a read voltage level to be provided to a word line, a sensing operation may be performed the same as a sensing operation using a read voltage decreased, by increasing a develop time.

(68) Accordingly, sensing the sensing node SO a plurality of times at a point in time when a develop time is changed may be the same as changing a word line voltage and precharging and sensing a bit line.

(69) FIG. 8A illustrates an example method 800 of performing a read operation on a memory device, according to some embodiments. FIG. 8A can be described with regard to elements of FIGS. 1-7. The example 800 can be performed by a storage system (for example, the storage system 10 of FIG. 1). It should be appreciated that the process flow 800 is not exhaustive and that other operation steps can be performed as well before, after, or between any of the illustrated operation steps. In some embodiments, some operation steps of the process flow 800 can be omitted or other operation steps can be included, which are not described here for simplicity. In some embodiments, operation steps of the process flow 800 can be performed in a different order and/or vary.

(70) At step 802, a first read operation is performed on memory cells (e.g., memory cell 608) in a memory device (e.g., memory device 100). In some examples, the first operation may include applying a pre-determined read voltage (e.g., V.sub.R15 in FIG. 5) to determine a state (e.g., P15) of one or more memory cells. In some aspects, as shown in diagram 520, the threshold voltage V.sub.th distribution of each state of P1-P15 may shift over time.

(71) At step 804, a first plurality of memory cells in a first state and a second plurality of memory cells in a second state are selected from the memory cells based on the first read operation. In some examples, a threshold voltage of each of the first plurality of memory cells in the first state may be larger than a read voltage in the first read operation and a threshold voltage of each of a second plurality of memory cells in the second state may be smaller than a read voltage in the first read operation. In some examples, the first state of a memory cell may correspond to a logical value of 1 in a sense latch. In some examples, a second state of a memory cell may correspond to a logical value of 0 in a sense latch. In some examples, one or more bit lines or sensing nodes associated with the first plurality of memory cells may be precharged after the first read operation. One or more bit lines or sensing nodes associated with the second plurality of memory cells may not be

precharged after the first read operation.

(72) At step **806**, a second read operation is performed on the first plurality of memory cells including the steps **816**, step **826** and step **836**. At step **816**, first data from the first plurality of memory cells through a first group of bit lines (e.g., bit line **618**) are sensed during a first develop time. In some examples, the first develop time may be associated with the develop period and the sense period as described with reference to FIG. 7. At step **826**, first data from the first plurality of memory cells through a second group of bit lines (e.g., bit line **628**) are sensed during a second develop time. At step **836**, first data from the first plurality of memory cells through a third group of bit lines (e.g., bit line **638**) are sensed during a third develop time. In some examples, the first develop time may include a time period, such as “T”, the second develop time may include a time period that is twice as the first develop time, such as “2T” and the third develop time may include a time period that is three times as the first develop time, such as “3T”. In some examples, the number of develop time may not be limited to three, and may include any number. In some examples, the number of develop time may be selected based on a different number of groups of the bit lines.

(73) In some examples, a first discharge path (e.g., SO discharge path **654**) associated with the first group of bit lines may be disabled after the first develop time. In some examples, a second discharge path (e.g., SO discharge path **664**) associated with the second group of bit lines may be disabled after the second develop time. In some examples, a third discharge path (e.g., SO discharge path **674**) associated with the third group of bit lines, may be disabled after the third develop time, to thereby perform the second read operation on the first plurality of memory cells.

(74) In some examples, the first data corresponding to each of the first plurality of memory cells are latched based on the second read operation to a logical value of one or zero by one or more sense latches (e.g., sense latch **656**, **666**, and **676**). The first data may be transferred to and modified in latch **680**. A modified logical value corresponding to each of the first plurality of memory cells may be obtained by latch **680**. In some examples, a logical value of one may be modified to a logical value of zero. Alternatively, a logical value of zero may be modified to a logical value of one.

(75) At step **808**, a third plurality of memory cells is selected, from the first plurality of memory cells, based on the second read operation. In some examples, the third plurality of memory cells from the first plurality of memory cells may be selected, based on the modified logical value. In some examples, the modified logical value corresponding to each of the third plurality of memory cells may be one. In some examples, the modified logical value corresponding to each of the third plurality of memory cells may be a predetermined value.

(76) At step **810**, a third read operation on the third plurality of memory cells is performed including step **820**, and step **830**. At step **820**, second data from the third plurality of memory cells through the second group of bit lines are sensed during a fourth develop time. At step **830**, second data from the third plurality of memory cells through the third group of bit lines are sensed during a fifth develop time. In some examples, the fourth develop time is shorter than the second develop time and the fifth develop time is shorter than the third develop time. In some examples, a first difference between the second develop time and the first develop time, and a second difference between the third develop time and the second develop time, may be the same. In some examples, a third difference between the second develop time and the fourth develop time, and a fourth difference between the third develop time and the fifth develop time, may be the same. In some examples, the third difference and the fourth difference may be the same as the first develop time. In some examples, the fourth develop time may be equal to the first develop time. In some examples, the fifth develop time may be equal to the second develop time.

(77) In some examples, a second discharge path associated with the second group of bit lines may be disabled after the fourth develop time; a third discharge path associated with the third group of bit lines may be disabled after the fifth develop time, to thereby perform the third read operation on

the third plurality of memory cells.

(78) In some examples, the second data may be latched corresponding to each of the third plurality of memory cells based on the third read operation to a logical value of one or zero.

(79) In some examples, a first number of memory cells associated with the first group of bit lines corresponding to a logical value of one may be determined. A second number of memory cells associated with the second group of bit lines corresponding to a logical value of one may be determined. A third number of memory cells associated with the third group of bit lines corresponding to a logical value of one may be determined.

(80) At step **812**, a read develop time is determined based on the third read operation.

(81) In some examples, the first number, the second number and the third number may be compared to obtain a minimum number of the first number, the second number and the third number. One of the first group of bit lines, the second group of bit lines, and the third group of bit lines corresponding to the minimum number of the first number, the second number and the third number may be determined. A develop time associated with the determined one of the first group of bit lines, the second group of bit lines, and the third group of bit lines may be determined as a read develop time. The first group of bit lines corresponding to the minimum number of the first number, the second number and the third number may be determined. A first develop time associated with the first group of bit lines may be determined as a read develop time. The read develop time may be an optimal develop time to determine one data state (e.g., **P15**) of a memory device, to minimize the impacts of threshold voltage (V_t) distribution shift using a pre-defined read level (e.g., $V_{sub.R15}$) in FIG. 5.

(82) In some examples, the first group of bit lines may include a first number of bit lines. In some examples, the second group of bit lines may include a second number of bit lines. In some examples, the third group of bit lines may include a third number of bit lines. In some examples, the first number, the second number, and the third number may be the same.

(83) In some examples, the fourth develop time and the first develop time may be the same; and the fifth develop time and the second develop time may be the same.

(84) FIG. **8B** illustrates an example threshold voltage distribution and data structures during the process flow of FIG. **8A**, according to some embodiments. FIG. **8B** can be described with regard to FIG. **8A** and FIGS. **1-7**. Not all of the depicted components may be used, however, and one or more implementations may include additional components not shown in the figure. Variations in the arrangement and types of the components may be made without departing from the spirit or scope of the claims as set forth herein. Additional, different or fewer components may be provided.

(85) FIG. **8B** includes a first data structure **892**, a second data structure **896** and a third data structure **890**.

(86) In some examples, first data structure **892** may include data associated with a latch (e.g., **680**) in a page buffer (e.g., page buffer **52**) associated with one or more memory cells, during the first read operation at step **802** in FIG. **8A**. For example, first data structure **892** may be associated with page buffer **52** in FIG. **6B**. First data structure **892** may include D_S_g0 , D_S_g1 , D_S_g2 , D_S_g3 , and D_S_g4 , corresponding to a first bit line group **g0**, a second bit line group **g1**, a third bit line group **g2**, a fourth bit line group **g3**, and a fifth bit line group **g4**. Each of D_S_g0 , D_S_g1 , D_S_g2 , D_S_g3 , and D_S_g4 may include logical value of 1 or 0 to indicate a first state or a second state of the memory cells.

(87) As described with reference to FIG. **6B**, during a latch period, a state of **SO 650** associated with the selected memory cell may be latched to sense latch **656** as a first state of the selected memory cell with a logical value of 1. Alternatively, a state of **SO 650** associated with the selected memory cell may be latched to sense latch **656** as a second state of the selected memory cell with a logical value of 0. In some examples, a threshold voltage of the selected memory cell in the first state is larger than a read voltage in the read operation and a threshold voltage of the selected memory cell in the second state is smaller than a read voltage in the read operation. After the latch

period, latch **680** may store the data associated with each of sense latch **656**, sense latch **666**, and sense latch **676**, as data related to the bit lines, such as the first data structure **892**.

(88) In some examples, second data structure **896** may include data associated with one or more sense latches (e.g., sense latch **656**, **666**, and **676**) and latch (e.g., **680**) in a page buffer (e.g., page buffer **52**) associated with one or more memory cells, during the second read operation at step **806**. As described above, the second read operation is performed on the first plurality of memory cells including the steps **816**, step **826** and step **836**. Second data structure **896** may include data associated with five sense latches SO_g0, SO_g1, SO_g2, SO_g3, and SO_g4. Each of the SO_g0, SO_g1, SO_g2, SO_g3, and SO_g4 may correspond to a sense latch associated with first bit line group g0, second bit line group g1, third bit line group g2, fourth bit line group g3, and fifth bit line group g4. The sense latch may be or include sense latch **656**, **666** or **676**.

(89) In some examples, second data structure **896** includes second data structure **896A**, **896B** and **896C**. Second data structure **896A** may include data associated with one or more sense latches associated with the first plurality of memory cells. Second data structure **896A** may include data associated with one or more sense latches associated with the first plurality of memory cells. Second data structure **896B** may include data associated with one or more sense latches associated with the first plurality of memory cells, after the steps **816**, step **826** and step **836** in step **806**. Second data structure **896C** may include modified data associated with latch (e.g., **680**) in a page buffer (e.g., page buffer **52**) associated with the first plurality of memory cells, after the steps **816**, step **826** and step **836** in step **806**. Latch in the page buffer may modify the data associated with the sense latches SO_g0, SO_g1, SO_g2, SO_g3, and SO_g4 to modify the logical value of 1 to the logical value of 0 or vice versa, as described with reference to FIG. **6B** and FIG. **8A**. Latch in the page buffer may provide the modified data in second data structure **896C**.

(90) In some examples, third data structure **890** may include data associated with one or more sense latches (e.g., sense latch **656**, **666**, and **676**) and latch (e.g., **680**) in a page buffer (e.g., page buffer **52**) associated with one or more memory cells, during the third read operation at step **810**. As described above, the third read operation is performed on the third plurality of memory cells including step **820**, and step **830**.

(91) Third data structure **890** includes third data structure **890A**, **890B** and **890C**. In some examples, third data structure **890A** may include data associated with one or more sense latches associated with the third plurality of memory cells. In some examples, third data structure **890B** may include data associated with one or more sense latches associated with the third plurality of memory cells, after step **820**, and step **830** in step **810**. In some examples, third data structure **890C** may include data associated with latch (e.g., **680**) in a page buffer (e.g., page buffer **52**) associated with the third plurality of memory cells, after step **820**, and step **830** in step **810**.

(92) In some examples, as described above with reference to step **812**, a read develop time is determined based on the third data structure **890C**.

(93) In some examples, one or more cell numbers in a first state or a second state associated with first data structure **892**, second data structure **896** and third data structure **890**, may correspond to one or more regions in the exemplary threshold voltage distribution shown in FIG. **8B**.

(94) In the present disclosure, a read develop time is determined based on a third read operation, after a first read operation and a second read operation. The read develop time may be an optimal develop time to determine one data state (e.g., **P15**) of a memory device, to minimize the impacts of threshold voltage (Vt) distribution shift using a pre-defined read level (e.g., **VR15**) in FIG. **5**.

(95) Among others, two advantages with the present disclosure can include: (1) an improved read efficiency can be performed by obtaining the Vt distribution of multiple regions in the threshold voltage distribution at once, which saves not only the read sensing time but also the cell number counting time; (2) an improved system design can be performed by only one latch (e.g., latch **680**) in the page buffer, thereby improving one or more circuit designs and performances.

(96) The foregoing description of the specific embodiments will so fully reveal the general nature

of the present disclosure that others can, by applying knowledge within the skill of the art, readily modify and/or adapt, for various applications, such specific embodiments, without undue experimentation, and without departing from the general concept of the present disclosure. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the disclosure and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the disclosure and guidance.

(97) Embodiments of the present disclosure have been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

(98) The Summary and Abstract sections can set forth one or more but not all exemplary embodiments of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the present disclosure and the appended claims in any way.

(99) The breadth and scope of the present disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A memory device, comprising: a memory array having memory cells; a page buffer coupled to the memory array through bit lines, wherein the page buffer comprises a latch; and a control logic coupled to the page buffer and configured to: perform a first read operation on the memory cells; perform a second read operation on the memory cells by dividing the memory cells into a plurality of groups of memory cells, and performing a second read operation on the plurality of groups of memory cells based on a first set of develop times, wherein each of the first set of develop times is different; perform a third read operation on the memory cells by performing a third read operation on the plurality of groups of memory cells based on a second set of develop times, wherein each of the second set of develop times is different; and determine a read develop time based on the third read operation.

2. The memory device of claim 1, wherein the control logic is further configured to: select, from the memory cells, a first plurality of memory cells in a first state and a second plurality of memory cells in a second state, based on the first read operation; and select, from the first plurality of memory cells, a third plurality of memory cells based on the second read operation.

3. The memory device of claim 2, wherein to perform the second read operation on the memory cells, the control logic is further configured to: sense first data from the first plurality of memory cells through a first group of bit lines based on a first develop time; sense first data from the first plurality of memory cells through a second group of bit lines based on a second develop time; and sense first data from the first plurality of memory cells through a third group of bit lines based on a third develop time; to perform the third read operation on the memory cells, the control logic is further configured to: sense second data from the third plurality of memory cells through a second group of bit lines during a fourth develop time; and sense second data from the third plurality of memory cells through a third group of bit lines during a fifth develop time, the fourth develop time being shorter than the second develop time, the fifth develop time being shorter than the third develop time; and the first set of develop time comprises the first develop time, the second develop time, and the third develop time, and the second set of develop time comprises the fourth develop time and the fifth develop time.

4. The memory device of claim 1, wherein a number of the plurality of groups is 3 to 5.

5. The memory device of claim 2, wherein a threshold voltage of each of the first plurality of memory cells in the first state is larger than a read voltage in the first read operation, and a threshold voltage of each of a second plurality of memory cells in the second state is smaller than a read voltage in the first read operation.
6. The memory device of claim 3, wherein the control logic is further configured to: disable a first discharge path associated with the first group of bit lines after the first develop time; disable a second discharge path associated with the second group of bit lines after the second develop time; and disable a third discharge path associated with the third group of bit lines after the third develop time to thereby perform the second read operation on the first plurality of memory cells.
7. The memory device of claim 3, wherein the control logic is further configured to: latch the first data corresponding to each of the first plurality of memory cells based on the second read operation to a logical value of one or zero; modify the latched first data by obtaining a modified logical value corresponding to each of the first plurality of memory cells; and select the third plurality of memory cells from the first plurality of memory cells, wherein the modified logical value corresponding to each of the third plurality of memory cells is one.
8. The memory device of claim 3, wherein the control logic is further configured to: disable a second discharge path associated with the second group of bit lines after the fourth develop time; and disable a third discharge path associated with the third group of bit lines, after the fifth develop time, to thereby perform the third read operation on the third plurality of memory cells.
9. The memory device of claim 3, wherein the control logic is further configured to: latch the second data corresponding to each of the third plurality of memory cells based on the third read operation to a logical value of one or zero; and determine a first number of memory cells associated with the first group of bit lines corresponding to a logical value of one, a second number of memory cells associated with the second group of bit lines corresponding to a logical value of one, and a third number of memory cells associated with the third group of bit lines corresponding to a logical value of one.
10. The memory device of claim 9, wherein the control logic is further configured to: compare the first number, the second number, and the third number to obtain a minimum number of the first number, the second number, and the third number; determine one of the first group of bit lines, the second group of bit lines, and the third group of bit lines corresponding to the minimum number of the first number, the second number, and the third number; and determine a read develop time, the read develop time comprising a develop time applied on a group of bit lines corresponding to the minimum number during the second read operation or the third read operation.
11. The memory device of claim 3, wherein the first group of bit lines comprises a first number of bit lines; the second group of bit lines comprises a second number of bit lines; the third group of bit lines comprises a third number of bit lines; and the first number, the second number, and the third number are the same.
12. The memory device of claim 3, wherein a first difference between the second develop time and the first develop time and a second difference between the third develop time and the second develop time are the same.
13. The memory device of claim 3, wherein the fourth develop time and the first develop time are the same, and the fifth develop time and the second develop time are the same.
14. The memory device of claim 7, wherein to modify the latched first data, the control logic is further configured to modify the latched first data with the logical value of one to a logical value of zero, and modify the latched first data with the logical value of zero to a logical value of one.
15. A method for operating a memory device, comprising: performing a first read operation on memory cells in the memory device; performing a second read operation on the memory cells, comprising dividing the memory cells into a plurality of groups of memory cells, and performing a second read operation on the plurality of groups of memory cells based on a first set of develop times, wherein each of the first set of develop times is different; performing a third read operation

- on the memory cells, comprising performing a third read operation on the plurality of groups of memory cells based on a second set of develop times, wherein each the second set of develop times is different; and determining a read develop time based on the third read operation.
16. The method of claim 15, further comprising: selecting, from the memory cells, a first plurality of memory cells in a first state and a second plurality of memory cells in a second state, based on the first read operation; and selecting, from the first plurality of memory cells, a third plurality of memory cells based on the second read operation.
17. The method of claim 16, wherein performing a second read operation on the memory cells comprises: sensing first data from the first plurality of memory cells through a first group of bit lines based on a first develop time; sensing first data from the first plurality of memory cells through a second group of bit lines based on a second develop time; and sensing first data from the first plurality of memory cells through a third group of bit lines based on a third develop time; performing a third read operation on the memory cells comprises: sensing second data from the third plurality of memory cells through a second group of bit lines during a fourth develop time; and sensing second data from the third plurality of memory cells through a third group of bit lines during a fifth develop time, the fourth develop time being shorter than the second develop time, the fifth develop time being shorter than the third develop time; and the first set of develop time comprises the first develop time, the second develop time, and the third develop time, and the second set of develop time comprises the fourth develop time and the fifth develop time.
18. The method of claim 15, wherein a number of the plurality of groups is 3 to 5.
19. A memory system, comprising: a memory device, comprising: a memory array having memory cells; a page buffer coupled to the memory array through bit lines, wherein the page buffer comprises a latch; and a control logic coupled to the page buffer and configured to: perform a first read operation on the memory cells; perform a second read operation on the memory cells by dividing the memory cells into a plurality of groups of memory cells, and performing a second read operation on the plurality of groups of memory cells based on a first set of develop times, wherein each of the first set of develop times is different; perform a third read operation on the memory cells by performing a third read operation on the plurality of groups of memory cells based on a second set of develop times, wherein each of the second set of develop times is different; and determine a read develop time based on the third read operation; and a memory controller coupled to the memory device and configured to control the memory device.
20. The memory system of claim 19, wherein a number of the plurality of groups is 3 to 5.
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